Customer Churn Analytics

Executive brief | Decision support for a retention pilot | Prepared by Haylynn Deepshika

Executive decision

Run a small, controlled retention pilot focused first on customers with repeated support needs and short tenure. Use the risk score to prioritise human review—not to automatically classify customers or issue discounts.

Metric	Observed result	Definition
Customers	15	Synthetic sample records
Churn rate	46.7%	7 churned ÷ 15 customers
Retention rate	53.3%	8 retained ÷ 15 customers
Monthly charges at risk	■409.93	Sum of monthly charges for churned records
Average tenure	18.8 months	Mean tenure across all records

Five decision-relevant insights

• **1. Churn is a visible issue in this sample.** 7 of 15 customers are marked churned (46.7%). This is a sample description, not an estimate of the wider customer base.

• **2. Churned records represent ■409.93 in monthly charges.** This is a simple monthly revenue-exposure proxy. It is not lifetime value, profit, or a forecast.

• **3. Contract type should be reviewed.** Month-to-Month accounts have 7 churned out of 7 in this sample. Small group sizes make the rate unstable.

• **4. Support experience is a useful review signal.** Average support tickets are 4.3 for churned records versus 1.0 for retained records. This association does not prove that tickets cause churn.

• **5. The segmentation creates actionable review groups.** The largest groups are New & High Support (5), Developing Customers (4), Established & Stable (4), Month-to-Month Watchlist (2). These are rule-defined cohorts, not discovered natural clusters.

Recommended action

• **Priority 1 — New & High Support:** contact customers with short tenure and repeated tickets; resolve open issues and offer onboarding support.

• **Priority 2 — Month-to-Month Watchlist:** ask why customers prefer monthly billing; explain annual-plan value and offer an optional incentive only where appropriate.

• **Priority 3 — Developing Customers:** run a short satisfaction check-in and monitor support signals.

• **Protect stable customers:** maintain service quality and recognise loyalty without unnecessary discounting.

Expected impact: scenario, not a promise

If a pilot were able to retain 10%–25% of the monthly charges currently associated with churned records, the illustrative retained monthly-charge range would be approximately **■40.99–■102.48 per month** for this sample. This is a sensitivity scenario calculated as ■409.93 × 10% to 25%; it is not a forecast, causal estimate, or guaranteed outcome. Actual impact depends on customer eligibility, intervention cost, conversion, margin and sustained retention.

Measurement plan

Run a time-limited pilot with a comparable holdout group if feasible. Track contact rate, issue-resolution rate, conversion to retained status, incremental monthly recurring charges retained, discount cost and customer complaints. Compare outcomes with the holdout before claiming impact.

Limitations and assumptions

• Only 15 synthetic customer records are available; results are not statistically generalisable.

• The rule-based score uses contract type (+3), tenure ≤12 months (+2), support tickets ≥4 (+2) and monthly charges ≥■80 (+1). Low = 0–2, Medium = 3–4, High = 5–8. It is not a probability of churn.

• The observed Churn label was excluded from risk-score and segment assignment; it was used only for descriptive post-hoc comparison.

• Thresholds are illustrative and not statistically calibrated. The dataset lacks longitudinal usage, customer feedback, complaint severity, intervention history and profit/margin data.

• Revenue-at-risk assumes monthly charges are comparable recurring amounts and does not account for taxes, costs, discounts or lifetime value.

Dashboard and supporting work

Live dashboard: https://haylynndeeshika.github.io/customer-churn-data-quality-analysis/
Supporting notebooks, CSVs, risk scores, memo and presentation are in the public GitHub repository: customer-churn-data-quality-analysis.

Conclusion: Use the analysis to focus a carefully measured retention pilot. Validate on larger real-world data before operational deployment.